

PetCuesGame – Educational Animal Safety Game*

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Abstract—The PetCuesGame is an interactive education game delivered via the web that is aimed at educating children on safe and respectful pet interaction. The project fills a critical need in conventional animal safety education, which too frequently consists of static flyers and live presentations that are boring and of little long-term recall. By using option-based gameplay and interactive scenarios, the game introduces an interactive world where children (ages 5-12) can make choices and get instant feedback on correct pet interaction behaviors.

In alignment with the United Nations Sustainable Development Goals (SDGs), the project supports Good Health and Well-being (SDG 3) through decreased risk of animal-related injuries, Quality Education (SDG 4) via an accessible, gamified educational platform, and Life on Land (SDG 15) through empathy and responsible pet ownership. The game is developed with the Godot Engine (GDScript) and is optimized for PC, tablet, and mobile use, making it accessible to a wide audience.

This paper presents the PetCuesGame's technical architecture, pedagogical approach, and local and international potential. With an innovative and scalable solution, the project not only enhances the Regina Humane Society's responsible pet ownership mandate, but also lays the groundwork for future development, such as multilingualism and additional animal behavior scenarios. The paper recapitulates the problem, solution, and vision for a safer, more knowledgeable community through creative digital learning.

I. INTRODUCTION

Background – Why is animal safety education important?

Animal safety education is a significant component in promoting safe and respectful human-animal interactions, particularly in children who are frequently exposed to pets both in public settings and within the home environment. With increased pet ownership across the globe, the likelihood of children encountering animals has significantly increased. Whereas such interactions can help in developing compassion and emotional maturity, their mishandling or misinterpretation of the behavior of animals can result in injury or aversive experiences for both the child and the animal. One of the foremost causes of dog bites and subsequent injury, according to animal welfare societies, is a deficiency of information on the proper approach and mode of communication with animals in a safe manner. Educating kids on reading animal body language and how to interact with them in a responsible manner minimizes the risk of an accident and makes it a

safer community for humans and pets alike. Additionally, educating kids on proper animal safety education teaches empathy, responsibility, and compassion, leading to healthier human-pet relationships.

Problem Statement – Lack of structured, interactive learning for safe pet interaction

Although education regarding animal safety is important, few structured and interactive materials adequately teach children how to interact with pets safely. Traditional educational materials such as brochures, classroom presentations, and live demonstrations fail to effectively hold the attention of young learners or provide them with the decision-making practice opportunity they need to remember. This education deficiency increases not only the risk of improper pet contact but also limits the ability to educate youth in responsible pet ownership at an early age. Current animal safety programs are largely passive and inaccessible to a majority of communities, so there is a need for an interactive, scalable, and engaging learning tool that will effectively teach animal safety protocols.

Research Objectives – Educate children using an interactive, web-based approach

The primary objective of the PetCuesGame project is to design an interactive web-based educational game that will teach children (ages 5-12) about safe and respectful interactions with pets. The project bridges the current gap in animal safety education by offering real-world scenarios where the players engage with virtual animals and receive immediate feedback on their actions. The game seeks to:

PetCuesGame seeks to promote safe animal interactions by teaching children appropriate behaviors such as seeking permission to pet an animal first, recognizing animal body language, and how to approach and interact with animals appropriately and safely. Through choice-based gameplay, the game reinforces learning through interactivity, rendering it enjoyable, memorable, and captivating. Accessibility and scalability are at the heart of the project design, with a web-based system that is optimized to perform perfectly on PCs, tablets, and mobile devices. This allows for ease of adoption in varied environments, including schools, humane societies, and homes. Additionally, PetCuesGame is aligned with the United Nations Sustainable Development Goals (SDGs) by supporting SDG 3 (Good Health & Well-being) through the reduction of

animal-related injuries, SDG 4 (Quality Education) through the delivery of an innovative and interactive educational product, and SDG 15 (Life on Land) through the promotion of proper pet care and interaction.

By leveraging technology and interactive learning, PetCues-Game aims to transform the dissemination of animal safety education to be accessible, effective, and enjoyable for diverse communities. The manuscript explores the technological infrastructure, pedagogical worth, and worldwide potential of the project while analyzing the disadvantages and future opportunities for improving animal safety education.

II. LITERATURE REVIEW

A. Game-Based Learning and Its Impact on Motivation

Game-based learning (GBL) has been largely associated with enhanced learner motivation through the provision of participatory and interactive environments. Liao et al. (2009) argue that game-based virtual learning environments foster intrinsic motivation and motivation through facilitating the interaction of players with virtual others as they perform effort-making and identity-making activities. The authors suggest a Performance-Based Learning (PeBL) approach, with emphases on two broad dimensions—effort-making and identity-making. In the making-effort component, participants engage in back-stage behavior in which they habitually practice skills through training virtual pets. In contrast, the making-identity component involves front-stage performance, where students showcase progress, hence promoting self-regulation and increasing motivation through peer evaluation (Liao et al., 2009, p. 430). Similarly, She et al. (2013) demonstrate that social-network-based learning games on platforms like Facebook can sustain and increase students' motivation. Their experiment with the "Operating System Little Pet (OSLP)" game reveals that incorporating social elements—such as virtual pet fights and leaderboards—encourages students to be more interested in course content. The study found that the students were keen to use the game outside of regular class hours, and competitive elements highly motivated them (She et al., 2013, p. 226).

B. Virtual Pets as Educational Tools

Use of virtual pets as learning aids is an effective method to stimulate interest and facilitate learning. As suggested by Liao et al. (2009), Our-Stage and My-Pet provide learners with an opportunity to interact with virtual pets in a dual-mode learning framework. On the Our-Stage system, students train and maintain their pets through a series of problem-solving experiences, but the Our-Stage system enables students to expose their pets to performances and competitions (Liao et al., 2009, p. 430). This two-pronged strategy combines cognitive acquisition (via task accomplishment) with social motivation (via public display), setting up a balanced environment that promotes both skill building and affective involvement.

She et al. (2013) affirm this concept by showing how the OSLP game uses virtual pets to connect course content with immediate decision-making. Students in their study demonstrated increased interest and performance when allowed to

breed and compete virtual pets. The sociality created through pet battles and leaderboards enabled learning and caused students to relearn and master the material (She et al., 2013, p. 226).

C. Social Interaction and Collaborative Learning

Both studies promote social interaction in game-based learning. She et al. (2013) outline the concept that social-network-based games provide a collaborative learning experience where students interact, compete, or cooperate asynchronously. This has the flexibility that allows learners to learn at their own pace while interacting with material but still relate to their fellow learners through competitive elements such as leaderboards and pet battles. The study indicated that these social factors not only improved learning accomplishments but also sustained long-term engagement (She et al., 2013, p. 226).

Liao et al. (2009) extend this concept further by detailing how performance-based learning facilitates peer-to-peer critique and public recognition. In the Our-Stage system, students present their trained pets on a virtual stage where observers (other students) view, criticize, and provide feedback. This repeated cycle of performance and criticism creates self-reflection and motivation. Secondly, the study found that the public performance helps students internalize learning through observation and community-based learning (Liao et al., 2009, p. 430).

D. Relevance to the PetCuesGame Project

Implications from these findings are directly applicable to the design of PetCuesGame:

PetCuesGame is particularly geared towards promoting safe and respectful interaction with animals by teaching children about key behaviors such as asking permission to pet, interpreting animal body language, and introducing pets safely and appropriately. By involving choice, the game promotes active learning, making the experience enjoyable, interactive, and memorable. Its internet-based platform has accessibility and scalability, functioning beautifully on PCs, tablets, and smartphones—enabling effortless integration into schools, humane societies, and home environments. Furthermore, PetCuesGame aligns with several United Nations Sustainable Development Goals (SDGs), including SDG 3 (Good Health and Well-being) by preventing animal-caused injuries, SDG 4 (Quality Education) through offering an interactive digital educational tool, and SDG 15 (Life on Land) by promoting responsible pet care and ownership.

III. GAME DESIGN AND EDUCATION FRAMEWORK

A. Core Elements of Game-Based Learning Design

a) *Motivation and Engagement:* Learning games like PetCuesGame must be designed so that they foster intrinsic motivation by facilitating the formation of affective bonds by taking care of virtual pets. This interaction not only enhances engagement but also enhances learning, since learners become individually invested in the well-being of their virtual pets (Liao et al., 2009, p. 430). In addition, incorporating elements

of competition and social engagement, such as pet battles, achievements, and leaderboards, can motivate students to continue being active and return to the material repeatedly, resulting in greater long-term retention and participation (She et al., 2013, p. 226). To enable goal-oriented learning, the game should also enable effective knowledge transfer by enabling players to achieve goal-oriented missions in the presence of explicit goals and realistic missions simulating, for example, meeting safe approaches to and animal handling—appropriate abilities directly beneficial to real life.

b) Feedback and Assessment: Continuous feedback is a key component of effective educational game design. Research points out that real-time responses to player moves allow learners to identify and correct mistakes on the spot, reinforcing learning and streamlining the learning process (She et al., 2013, p. 225). Additionally, embedding formative assessment through in-game quizzes, interactive exercises, or task-based checkpoints allows learners to track their ongoing progress. These assessments offer not only several opportunities for practice but also responsiveness to the needs of individual learners, promoting mastery at an individual pace (Liao et al., 2009, p. 431). Together, these methods allow for that gameplay remains interesting and learning-wise effective.

B. Game Narrative and Contextual Learning

Narrative-driven gameplay is an effective tool for enhancing knowledge retention within learning games. Placing learning within a motivating narrative context allows players to become emotionally engaged with the content, making educational exercises more meaningful and stimulating. Liao et al. (2009) pinpoint how performance-based learning within a virtual pet world creates an environment in which players are motivated to engage in activities that are educational and enjoyable (p. 430). In addition, making real-world situations part of the gameplay enhances depth in the learning process. According to She et al. (2013), presenting course material in interactive and realistic situations makes it easier for students to grasp and apply the knowledge outside the game environment (p. 226). The combination of narrative simulation and applicability to the real world ensures that the skill learning within the game is sustainable and transferrable to actual situations.

In PetCuesGame:PetCuesGame uses story-driven gameplay to immerse children in contextual learning environments. Players are the helpers, guiding lost animals back home or riding along with animals as they travel through the neighborhood. Story-driven missions use empathy while situating learning safety skills within context. For example, when a child is confronted with the scenario of finding a lost dog, they are naturally prompted to act responsibly and safely, such as selecting to "Ask an adult for help." The game incorporates such real-life situations in an enjoyable and engaging narrative format and encourages critical thinking as well as learning safe practices in a realistic and memorable experience for small children.

C. Impact on Stakeholders

The PetCuesGame project has been designed with the emphasis on its key stakeholder, Ms. Rebecca Carson, the Regina Humane Society spokesperson. As a key humane education and animal welfare outreach representative of Saskatchewan, her commitment ensures the quality of the educational content and functional practicality of the game material. The integration of the game into existing outreach programs of the Regina Humane Society guarantees its potential for long-term behavioral influence, particularly in children, the main consumers of this educational tool.

From a stakeholder impact perspective, the game fulfills several basic needs:

1. Improved Humane Education Delivery: The Regina Humane Society benefits from an online resource that supports traditional education initiatives. By employing gamified learning, PetCuesGame offers an interactive, participatory experience that deconstructs complex concepts about pet safety and behavior into simpler terms for kids.

2. Enhanced Community Safety: The game assists children in identifying animal body language and approach strategies, which can help lower incidents involving pets and stray animals. This aligns directly with the Society's goal of creating safer and more educated communities.

3. Ease of Access to Educators and Parents: The game can be incorporated into classwork or animal safety lessons with no special resources or training involved. Parents can also rest easy knowing their children are receiving essential information from a credible, well-structured source.

4. Reputational and Outreach Amplification: Having Ms. Carson and the Regina Humane Society on board, the project gains more credibility and alignment with humane best practices. Additionally, the game provides measurable outreach data in user engagement levels and scenario completions to be used in future Society reports or educational program evaluations.

By positioning Rebecca Carson as a key stakeholder, and designing the PetCuesGame to operate both as an educational and animal welfare tool, the project pays back more than entertainment. It becomes a model of community-focused, mission-driven educational technology that facilitates safe, respectful human-animal relationships.

D. Game Scenario Development

Dog and child will interact with each other as shown below(Fig. 1), and child will select a correct option for the given Question.

In image below(Fig. 2, 3 and 4), we can see some of the pets' movements based on the question so the child can look at that and answer the question.

After the 15 questions system will give a score out of 15(Fig. 5).

IV. COMPARISON WITH EXISTING GAMES

Modern Educational Animal Safety Games Some of the educational games focus on pet safety or animal interaction



Fig. 1. Interaction



Fig. 2. Behavior

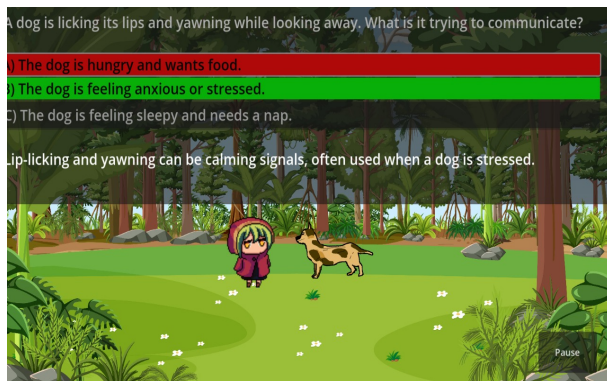


Fig. 3. Wrong Answer screen



Fig. 4. Behavior



Fig. 5. Score board

but are not interactive or do not make use of choice-based learning to a high standard. Those below have been analyzed:

Several existing educational games address safety and animal topics, but none of them provide the total and engaging solution that PetCuesGame offers. For instance, the Zoe and Molly Online Safety Game is a good educator on internet safety for kids but doesn't carry over to physical interaction with animals and thus is of no use for real pet safety. Doggone Crazy! as a board game and computer game educates about dog safety by means of cards and trivia, but it does not create a fun, interactive computer setting in which players can make real-time decisions. Fetch! With Ruff Ruffman, a PBS Kids game, includes pet-related activities but does not provide an in-depth, interactive curriculum dedicated to safety education. At the same time, Animal Jam teaches players about wildlife and environmental conservation with minimal attention to domestic pet safety. In contrast, PetCuesGame bridges this critical gap by incorporating systematic learning, interactive decision-making, and realistic pet behavior simulations.

Primary Differences and Advantages of PetCuesGame In contrast to other games, PetCuesGame offers a thorough, interactive learning experience through:

PetCuesGame is different from other games in that it offers an immersive, decision-based learning experience beyond the static presentation of information presented in games such as Doggone Crazy! and Animal Jam. In PetCuesGame, players themselves make decisions and assess their repercussions in real time, increasing participation and learning outcomes. The game also enables behavioral AI through finite state machines (FSMs) and animation to allow realistic, dynamic pet interaction over the predetermined, limited-situation scripts found in some other educational games. While TV programs like Fetch! With Ruff Ruffman feature pets as part of content, these do not consist of a structured curriculum of animal safety instruction. PetCuesGame fills this gap by offering an articulate, pedagogically sound system. Furthermore, the game is scalable and accessible, playable on multiple devices without needing special hardware—appropriate for widespread use in schools and humane societies. Nearly as important, the game aligns with the United Nations Sustainable Development

Goals (SDGs), namely SDG 3 (Good Health and Well-being), SDG 4 (Quality Education), and SDG 15 (Life on Land), through the promotion of responsible pet ownership and injury prevention.

TABLE I
TABLE TYPE STYLES

Feature	PetCue Games	Zoe& Molly	Doggone Crazy!	Fetch! with Ruff Ruffman
Interactive Scenarios	Yes	No	No	No
Real-Time Decision Making	Yes	No	No	No
Pet Behavior Simulation	Yes	No	No	No
Focus on Pet Safety	Yes	No	Yes	No
Gamified Learning	Yes	Yes	Yes	Yes
Cross-Platform Accessibility	Yes	No	No	Yes

^aComparison table

V. IMPLEMENTATION AND TECHNICAL ARCHITECTURE

The PetCuesGame is coded with the Godot Engine, an open-source game engine that has built-in support for GDScript, a Python-like language that is tailored for game logic. The game is designed as several interactive scenarios that mimic actual pet interactions in real life, giving children choice-based gameplay.

A. Core Game Components

a) User Interface (UI): The game has a user-friendly UI that leads students through various animal safety situations. The UI layout makes the content accessible to young students, using large buttons, voice-over instruction, and minimal text to hold their interest.

b) Character and Pet Animations: Personified animated pet characters with lifelike behavior copy real-world behaviors. Animation speech such as "happy," "scared," "aggressive," and "neutral" provides contextual feedback based on player input.

c) Dialogue System: A branch dialogue system is employed to provide dynamic interaction in which players can ask for permission, read pet body language, and receive instantaneous feedback.

d) Decision-Based Learning: The game employs decision trees where player choices influence the result. Positive reinforcement is given if the action is appropriate, and wrong choices trigger educational messages.

e) AI Behavior Simulation: The AI for pet behavior responds to moves by players, similar to real pets responding when subjected to different interactions.

B. Platform Compatibility

The game is cross-platform-optimized for smooth compatibility on:

- Web browsers (HTML5 export for easy access in educational settings)
- Tablets and mobile devices
- Desktop computers

VI. CLIENT COLLABORATION AND FEEDBACK

The development of PetCuesGame was conducted in close collaboration with the Regina Humane Society, in order to ensure that the project remained faithful to real educational goals and community needs. Rebecca, an official with the organization and lead client for the project, provided ongoing input throughout the design process.

Rebecca signed up immense enthusiasm towards the concept and vision of the game, seeing it to possess the potential effectively to tackle issues of utmost importance related to pet safety and child education. She appreciated the emphasis on dynamic, choice-driven learning and concurred with the position of PetCuesGame as being capable of making a robust, positive contribution to educational outreach initiatives.

In her otherwise encouraging remarks, Rebecca made some very useful suggestions which were instrumental in enhancing the project. She proposed increased pet variety to increase relatability and interest among the players. She also emphasized introducing more animations so that pet interaction could feel more real and emotionally engaging with children.

Overall, Rebecca was satisfied with the development process, particularly the team's openness to hearing feedback and commitment to educational quality. Her feedback gives credence to client-centered design in educational technology, highlighting the importance of stakeholder input in creating effective, meaningful tools.

VII. CHALLENGES AND LIMITATIONS

A. Development Challenges

Development of PetCuesGame was challenging due to various reasons. One such major challenge was to strike a balance between learning content and ease of gameplay so that children could learn and play easily. It was crucial to keep the tone—learning but not inhibitive—intact. Technically too, it was difficult to have real-like pet behavior through finite state machines (FSMs) with maximum performance retained. Another challenge was the incorporation of feedback from community partners, Regina Humane Society, and iteratively refining the game without extending the development timeline. Incorporation of features such as dialogue systems and feedback mechanisms also needed careful scripting to avoid bugs and maintain child-friendly pacing.

B. Limitations

The game currently revolves around common family pets, i.e., dogs, and does not yet reach exotic pet learning situations like birds or reptiles. Additionally, PetCuesGame is focused primarily for web-based systems, thereby restricting usage by people without consistent internet access or newer devices. The second limitation is that few scenarios are given under the MVP; additional content expansion is needed to offer ample learning objectives.

VIII. CONCLUSION AND FUTURE WORK

PetCuesGame successfully delivers an interactive, engaging learning process that assists in increasing children's knowledge of safe pet interaction. Through gamification, real-life situations, and behavior AI, the game delivers a realistic and scalable solution to promote responsible pet ownership. The project closely aligns with SDG targets 3, 4, and 15, making it socially relevant as well as educationally beneficial.

Future growth is seen by the development team in several enhancements. These are expanding the scope to accommodate more animals and cultural settings, introducing new levels and missions, introducing analytics dashboards to parents and teachers, and exploring augmented reality (AR) features for increased engagement. Through overcoming current limitations and incorporating user feedback, PetCuesGame can be a powerful learning tool used globally by schools, humane societies, and public education initiatives.

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